Microcontroller Based Temperature Control of an Industrial Process Using Advanced Control Techniques

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Introduction to Temperature Control

- Temperature control is important in many industrial and domestic applications such as HVAC systems, incubators, chemical processes, and furnaces. Maintaining a stable temperature improves system efficiency, safety, and product quality. Conventional control methods like ON–OFF and PID controllers are widely used but may cause oscillations and slower response in complex systems.

- With the development of microprocessor and microcontroller technology, embedded systems are increasingly used for digital control applications. Microcontrollers can read sensor data, process it using control algorithms, and control actuators in real time.

- In this project, a Smart Temperature Control System Using Advanced Control Techniques is developed. A temperature sensor measures the system temperature and sends data to a microcontroller. The microcontroller processes this data using an advanced control algorithm to maintain the desired temperature. MATLAB is used to simulate and analyze the system performance.

03.

Project Objectives

Smart Sensor Integration

- Deploy temperature sensors for accurate environmental monitoring

Real-Time Microcontroller Interface

- Connect sensors to microcontrollers for **instant data acquisition**

Advanced Control Algorithms

- Utilize sophisticated techniques for **precise temperature regulation**

MATLAB Simulation & Analysis

- Model and evaluate system performance using MATLAB tools

04.

System Methodology

- **Temperature Sensing**

A temperature sensor such as LM35 is used to measure the temperature of the environment. The sensor converts temperature into an analog electrical signal.

- **Signal Conversion**

The analog signal from the sensor is converted into digital form using the ADC of the microcontroller.

- **Microcontroller Processing**

The microcontroller receives the digital temperature data and processes it using a control algorithm. The algorithm compares the measured temperature with the desired setpoint value.

- **Control Action**

Based on the control algorithm output, the microcontroller generates a control signal to regulate the temperature.

- **Actuator Control**

The control signal drives an actuator such as a heater or cooling fan through a relay or driver circuit to maintain the desired temperature.

05.

Software and Tools Used

MATLAB for System Simulation

Enables precise modeling and analysis of temperature control.

Simulink for Control Design

Visual block diagrams streamline control system development.

Embedded C & Arduino IDE

Microcontroller programming for real-time sensor and actuator control.